HALX: Privacy-Preserving Hybrid Attention-LSTM and XGBoost Framework for Proactive Threat Detection in Financial Networks

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***Abstract*—Financial institutions need intrusion detectors that are accurate and auditable and that do not themselves become a channel for leaking customer data. We present HALX, a framework for proactive threat detection in financial networks with three parts: a feature-token Attention-LSTM encoder, an XGBoost scoring head, and a privacy-preserving inference layer. The privacy layer enforces Mondrian k-anonymity on quasi-identifiers and releases the preprocessing statistics under (ε, δ)-differential privacy. Every alert carries a threat score, a severity tier and the most-attended features as an explanation. We compare HALX with seven baselines over three seeds on the official UNSW-NB15 split and on a CICIDS2017 subset, giving every method the same privatized inputs. On CICIDS2017, HALX reaches an F1-score of 99.58% at a 0.77% false-positive rate. This is on par with the strongest tree ensembles and gives 2–10$\times$ fewer false alarms than CNN-LSTM, BiLSTM and Transformer detectors. On UNSW-NB15, a train–test distribution shift holds every model near 89% F1 on average. There HALX matches XGBoost (89.11%), has the most stable false-positive rate across seeds, and detects the most attacks at a 1% false-alarm budget (86.9%, seed 0). Under this shift, 97.0% of its high-severity alerts are real attacks. The privacy layer (k=10, ε=1) costs at most 0.16 F1 points, and HALX scores thousands of flows per second on a CPU. Code and results are publicly available.**

***Index Terms*—Intrusion detection, financial cybersecurity, attention-LSTM, XGBoost, k-anonymity, differential privacy, explainable AI, PCI DSS, GDPR**

I. Introduction

Payment switches, core-banking platforms and the cloud services that host them are prime targets. The financial sector had the second-highest average breach cost of 17 industries in the IBM 2024 study, at US$6.08 million per incident [1], and banks face DDoS, ransomware, insider misuse and multi-stage intrusions [2], [3]. Signature-based intrusion detection systems (IDS) miss unseen attacks, and a reactive security operations centre (SOC) learns of an intrusion only after the damage is done. *Proactive* detection instead scores every flow as it is observed and gives analysts a ranked, explained set of alerts.

Recent ML-based IDS use increasingly elaborate deep models: CNN–LSTM hybrids [4], [5], attention-augmented recurrent networks [6] and transformers [7]–[9]. Deploying them in a regulated institution raises three issues that benchmark accuracy does not capture. *(i) Tabular reality.* Flow records are tabular, where gradient-boosted trees remain hard to beat [10], so a deep detector must be compared with a tuned boosted-tree baseline. *(ii) Privacy and compliance.* Flow features such as transferred volumes, session durations and TTL or window values act as *quasi-identifiers* of customers and hosts. Under GDPR Art. 25 (data protection by design) [11] and PCI DSS [12], the detector and what it stores must not become a new leakage channel. *(iii) Auditability.* Analysts need to know *why* a flow was flagged and how urgent it is [13].

We present **HALX** (Hybrid Attention-LSTM + XGBoost), which addresses all three. Our contributions are:

- A hybrid detector: a feature-token Attention-LSTM encoder learns a flow representation that an XGBoost head [14] scores together with the privatized features, yielding a threat probability and three severity tiers.
- A privacy-preserving inference layer with *formally stated* guarantees: Mondrian k-anonymity [15] on quasi-identifiers and (ε, δ)-DP preprocessing statistics with zCDP accounting [16].
- A reproducible evaluation on UNSW-NB15 [17] and CICIDS2017 [18] against seven baselines over three seeds, with ablation, privacy–utility, explanation and latency analyses. Code and results are released.[1]

II. Related Work

Deep and hybrid IDS. CNN–LSTM models combine convolutional feature extraction with recurrent modelling and are a common baseline on CICIDS2017 and UNSW-NB15 [4], [5]. Alashjaee [6] adds self-attention to a CNN–LSTM and shows in an ablation that the attention layer helps. Transformer-based detectors treat flow features as tokens [7], [9], [19]. FlowTransformer showed that the choice of input encoding and classification head matters more than the transformer variant [8], and BERT-style language models have also been adapted to traffic [20]. Closest to our work, Sajid et al. [21] use XGBoost and a CNN as feature selectors in front of an LSTM classifier. HALX reverses that order: the deep encoder produces the representation and the boosted trees make the

[1] https://github.com/mazharulmd/Threat-Detection-for-Financial-Systems

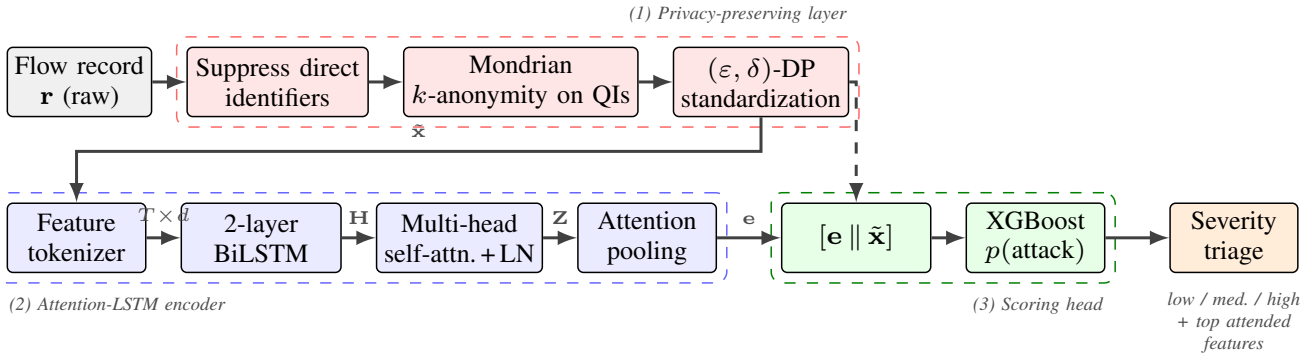


Fig. 1. The HALX pipeline. Raw flow records pass through the privacy layer (direct-identifier suppression, Mondrian k-anonymous generalization of quasi-identifiers, DP standardization). The Attention-LSTM encoder reads the privatized record as a sequence of feature tokens and produces an embedding $\mathbf{e}$. XGBoost scores $[\mathbf{e}\|\tilde{\mathbf{x}}]$, and the score is mapped to a severity tier together with the most-attended features.

decision. This follows evidence that trees are the stronger final learner on tabular data [10], [22].

ML security in finance. Surveys of digital banking identify DDoS, ransomware, insider threats and phishing as the dominant threat classes [2]. Islam et al. [3] detect DDoS on a banking IoT monitoring system. Recent studies report SOC deployments in core-banking systems [23] and proactive detection in emulated bank networks [24]. They stress regulatory constraints, but none of them states a formal privacy guarantee for the detector itself.

Privacy-preserving IDS. Most privacy work on IDS concerns *training*: federated learning [25] with differentially private updates [26]–[28], or DP-SGD [29]. We address a complementary problem that is often overlooked: what the deployed detector *exposes*, namely generalized quasi-identifiers in its inputs and logs and the normalization statistics shipped with the model. The two approaches can be combined.

Evaluation pitfalls and explainability. CICIDS2017 has labelling errors [30] and transfers poorly [31], and Arp et al. [32] warn against data snooping and weak baselines. We therefore never use test data for model selection, always include a tuned XGBoost, and report means over seeds. Unlike post-hoc SHAP explanations of XGBoost NIDS [13], [33], HALX explains alerts through its intrinsic attention weights.

III. Proposed Framework: HALX

Fig. 1 shows the three stages: (1) a privacy-preserving layer, (2) an Attention-LSTM encoder and (3) an XGBoost scoring head with severity triage. Algorithm 1 summarizes training.

Problem and threat model. A flow exporter in the institution's network summarizes each flow as a record with d numeric and c categorical attributes. External or insider attackers generate reconnaissance, exploit, DoS, brute-force, botnet and web-attack flows. For every flow HALX outputs a threat score $p \in [0,1]$, a decision $\hat{y} = \mathbb{1}[p \geq \tau]$, an attack family and a severity tier. An *honest-but-curious* operator or auditor who sees the stored inputs and released statistics must not be able to re-identify individual flows.

A. Privacy-Preserving Layer

Suppression. Direct identifiers (IP addresses, ports, timestamps, flow IDs) are dropped before any processing.

Quasi-identifier generalization. Let $Q \subset \{1,\dots,d\}$ be the quasi-identifiers (QIs). On UNSW-NB15 these are duration, source and destination bytes, and source TTL; on CICIDS2017 they are flow duration, forward and backward payload lengths, and the initial forward TCP window. Volumes play the role of transaction sizes, and TTL or window values fingerprint operating systems. After a public, data-independent transform $g(x) = \operatorname{sign}(x)\log(1+|x|)$, we build a strict Mondrian partition [15] of the training QIs. Each region is split recursively at the median of its widest normalized QI, as long as both children keep at least k records. Every record, in training or at inference, is routed to its leaf and its QIs are replaced by the leaf median. Hence every generalized QI vector the detector stores or logs is shared by at least k training records (k-anonymity [34]). We use $k = 10$ by default and verify the smallest equivalence class after fitting.

DP standardization. Standardization needs per-feature means and variances, which are normally shipped with a model and can reveal individual records when an outlier dominates. We release them with the Gaussian mechanism [35]. With values clipped to $[-C, C]$ ($C = 25$ in g-space) and public n, the sums $S_1 = \sum_i x_i$ and $S_2 = \sum_i x_i^2$ have replace-one sensitivities $2C$ and C^2. Each of the $2d$ released sums receives noise $\mathcal{N}(0, \Delta^2/2\rho_q)$ with $\rho_q = \rho/2d$, where

$$\rho = \left(\sqrt{\ln(1/\delta) + \varepsilon} - \sqrt{\ln(1/\delta)}\right)^2 \tag{1}$$

is the largest ρ for which ρ-zCDP implies (ε, δ)-DP, using $\varepsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}$ [16]. By composition, the released statistics satisfy (ε, δ)-DP. We use $\varepsilon = 1$ and $\delta = 10^{-5}$. *Scope:* this protects the shipped statistics only. The network weights are not differentially private; that would require DP-SGD [29] (Sec. VI).

B. Attention-LSTM Encoder

UNSW-NB15's partition files carry no timestamps and are sorted by label, and the CICIDS2017 subset is randomly sam-

Algorithm 1 HALX training

Require: training/validation flows $\mathcal{D}_{tr}, \mathcal{D}_{va}$; QI set Q; k, ε, δ

1: Drop direct identifiers; apply $g(\cdot)$; impute with training medians
2: Fit Mondrian tree on $\mathcal{D}_{tr}[Q]$ with leaf size $\geq k$; generalize Q
3: Release (μ, σ) via Gaussian mechanism with $\rho(\varepsilon, \delta)$ (Eq. 1); standardize $\rightarrow \tilde{\mathbf{x}}$
4: Oversample minority families in $\mathcal{D}_{tr}$ with SMOTE-NC [38]
5: Train encoder (Eqs. 2–4) with early stopping on $\mathcal{D}_{va}$
6: Extract $\mathbf{f} = [\mathbf{e} \| \tilde{\mathbf{x}}]$ for $\mathcal{D}_{tr}, \mathcal{D}_{va}$
7: Fit binary and multi-class XGBoost on $\mathcal{D}_{tr}$, early-stopped on $\mathcal{D}_{va}$
8: **return** privacy layer, encoder, XGBoost heads, threshold τ

pled. Sliding windows of *consecutive flows* would therefore leak labels through ordering, a form of data snooping [32]. HALX instead reads each flow record as a sequence of $T = d + c$ *feature tokens*, in the spirit of tabular transformers [22]:

$$\mathbf{t}_j = \tilde{x}_j \, \mathbf{w}_j + \mathbf{b}_j \;\; (j \leq d), \qquad \mathbf{t}_{d+i} = \mathbf{E}_i[r_i^{\text{cat}}] + \mathbf{b}_{d+i}, \quad (2)$$

with $\mathbf{w}_j, \mathbf{b}_j \in \mathbb{R}^{16}$. A two-layer bidirectional LSTM [36] with 32 units per direction produces $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 64}$. Here the recurrence models interactions between related features, for example packet counts, sizes and rates of the same direction. Multi-head self-attention [37] with $H = 4$ heads lets every token attend to every other:

$$\mathbf{Z} = \mathrm{LN}\big(\mathbf{H} + \mathrm{MHA}(\mathbf{H}\mathbf{W}^Q, \mathbf{H}\mathbf{W}^K, \mathbf{H}\mathbf{W}^V)\big), \quad (3)$$

where each head computes $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$. Additive attention pooling then yields the flow embedding together with an importance distribution over features:

$$\alpha_j = \mathrm{softmax}_j\big(\mathbf{u}^\top \tanh(\mathbf{W}_a \mathbf{z}_j)\big), \qquad \mathbf{e} = \textstyle\sum_j \alpha_j \mathbf{z}_j. \quad (4)$$

The weights α_j are reported with each alert as an explanation. The encoder is trained end to end with a linear softmax head on the multi-class labels (cross-entropy, AdamW, learning rate $3 \cdot 10^{-3}$, cosine schedule, batch 1024). Early stopping uses validation macro-F1.

C. XGBoost Scoring Head and Severity Triage

The deep encoder captures cross-feature structure, but on tabular data boosted trees model sharp, axis-aligned decision boundaries better [10]. The head therefore receives the concatenation $\mathbf{f} = [\mathbf{e} \,\|\, \tilde{\mathbf{x}}]$ of the embedding and the privatized features. One XGBoost model [14] estimates $p = P(\text{attack} \mid \mathbf{f})$ and a second predicts the attack family (up to 800 trees, depth 8, learning rate 0.08, early stopping on the validation split). Alerts with $p \geq \tau = 0.5$ are triaged into *low* ($p < 0.8$), *medium* ($0.8 \leq p < 0.95$) and *high* ($p \geq 0.95$) severity, so SOC analysts can work through high-severity alerts first.

IV. Implementation and Experimental Setup

A. Datasets and Protocol

UNSW-NB15 [17]: we use the official partition (175,341 training / 82,332 test flows; 9 attack families; 39 numeric and 3 categorical attributes, i.e. 42 tokens). 15% of the training part is held out for validation, and the test set is used only for final reporting. Its attack share differs from training (55.1% vs. 68.1%), which makes this the harder, more realistic setting. **CICIDS2017** [18]: we use the publicly released stratified subset of Yang et al. [39] (56,661 flows, 77 features, 6 grouped attack classes: DoS, PortScan, brute force, web attack, bot and infiltration; 60 informative numeric features). We split it 70/15/15 (39,662/8,499/8,500 flows) with stratification. Given known labelling issues [30], we treat it as a second, easier benchmark. Duplicate and constant columns (decided on training data only) are removed, infinite values are imputed with training medians, and categorical attributes keep their 31 most frequent levels plus an "other" bucket. SMOTE-NC [38] raises every attack family to at least 2% of the *training* split only. Every configuration is run with three seeds (0, 1, 2); the seed changes the train/validation split, SMOTE, weight initialization and the DP noise. We report mean $\pm$ std.

B. Baselines

All baselines receive the same privatized inputs: (i) Isolation Forest [40], trained on benign flows only, with the threshold chosen on validation F1; (ii) Random Forest [41] (300 trees); (iii) XGBoost [14], configured exactly like the HALX head; (iv) a two-layer MLP (256 units); (v) CNN-LSTM in the style of [4] (two Conv1d layers followed by an LSTM over feature tokens); (vi) the HALX BiLSTM without attention and without the XGBoost head; and (vii) a feature-token Transformer (FT-Transformer style [22], two layers, 4 heads, $d = 64$), which stands in for the transformer IDS family [7], [8]. The deep baselines use the same optimizer, schedule and early-stopping rule as HALX.

C. Implementation

HALX uses PyTorch 2, XGBoost 3 and scikit-learn; the privacy layer (Mondrian tree, zCDP standardizer) is $\approx$150 lines of NumPy. All runs used a 4-core x86 CPU (16 GB RAM, no GPU), at most 15 epochs (8 on UNSW-NB15) and patience 3. We report accuracy, precision, recall, F1 and FPR for the binary decision at $\tau = 0.5$, ROC-AUC, and macro-F1 (M-F1) over attack families.

V. Results

A. Detection Performance

Table I compares HALX with the seven baselines.

CICIDS2017. Every supervised tree-based method is close to the ceiling. HALX reaches F1 99.58% (FPR 0.77%), statistically level with Random Forest (99.57%) and within 0.1 points of XGBoost (99.68%). The pure deep models fall clearly behind: the Transformer reaches 99.15%, CNN-LSTM 98.04% and the attention-free BiLSTM 95.66%, with 2–10$\times$

TABLE I

BINARY DETECTION (ATTACK VS. BENIGN, $\tau = 0.5$) AND MULTI-CLASS MACRO-F1 (M-F1) ON THE TEST SETS, MEAN$\pm$STD OVER 3 SEEDS, IN %. BEST MEAN PER COLUMN AND DATASET IN BOLD. ALL METHODS RECEIVE IDENTICAL PRIVATIZED INPUTS ($k = 10$, $\varepsilon = 1$).

Method	Acc.	Prec.	Rec.	F1	FPR	AUC	M-F1
CICIDS2017 (stratified subset)							
Isolation Forest [40]	77.22$\pm$0.21	74.00$\pm$0.25	95.51$\pm$0.27	83.39$\pm$0.13	50.09$\pm$0.73	80.64$\pm$0.86	–
Random Forest [41]	99.48$\pm$0.08	99.59$\pm$0.03	99.55$\pm$0.11	99.57$\pm$0.07	0.62$\pm$0.04	99.88$\pm$0.03	**97.73$\pm$0.10**
XGBoost [14]	**99.62$\pm$0.08**	**99.63$\pm$0.08**	**99.74$\pm$0.06**	**99.68$\pm$0.06**	**0.56$\pm$0.13**	**99.96$\pm$0.01**	97.35$\pm$0.98
MLP	97.74$\pm$0.22	97.15$\pm$0.38	99.13$\pm$0.23	98.13$\pm$0.18	4.34$\pm$0.60	99.77$\pm$0.01	93.29$\pm$0.43
CNN-LSTM [4]	97.64$\pm$0.19	97.45$\pm$0.25	98.64$\pm$0.32	98.04$\pm$0.16	3.86$\pm$0.40	99.57$\pm$0.05	89.33$\pm$1.58
BiLSTM (no attention)	94.76$\pm$0.65	94.86$\pm$0.24	96.47$\pm$0.90	95.66$\pm$0.55	7.80$\pm$0.33	98.88$\pm$0.14	85.40$\pm$3.19
Transformer [22]	98.97$\pm$0.27	98.82$\pm$0.34	99.48$\pm$0.12	99.15$\pm$0.23	1.78$\pm$0.51	99.86$\pm$0.07	95.29$\pm$0.82
HALX (ours)	99.50$\pm$0.09	99.48$\pm$0.07	99.68$\pm$0.08	99.58$\pm$0.07	0.77$\pm$0.11	99.95$\pm$0.01	95.86$\pm$0.44
UNSW-NB15 (official split)							
Isolation Forest [40]	73.24$\pm$0.69	69.27$\pm$1.58	92.73$\pm$3.45	79.23$\pm$0.19	50.64$\pm$5.75	81.53$\pm$0.47	–
Random Forest [41]	86.67$\pm$0.09	81.18$\pm$0.10	**98.66$\pm$0.06**	89.07$\pm$0.07	28.03$\pm$0.20	97.86$\pm$0.02	49.04$\pm$0.31
XGBoost [14]	86.81$\pm$0.14	81.57$\pm$0.18	98.22$\pm$0.02	89.13$\pm$0.10	27.18$\pm$0.33	**98.26$\pm$0.03**	**49.93$\pm$0.84**
MLP	86.36$\pm$0.24	81.39$\pm$1.00	97.57$\pm$1.32	88.73$\pm$0.05	27.38$\pm$2.15	97.75$\pm$0.49	43.70$\pm$1.38
CNN-LSTM [4]	86.13$\pm$0.35	80.63$\pm$0.42	98.47$\pm$0.38	88.66$\pm$0.26	28.98$\pm$0.82	98.17$\pm$0.10	42.43$\pm$0.35
BiLSTM (no attention)	84.99$\pm$0.05	79.25$\pm$0.24	98.54$\pm$0.39	87.85$\pm$0.01	31.61$\pm$0.58	97.94$\pm$0.13	40.76$\pm$0.88
Transformer [22]	**86.94$\pm$2.60**	**82.28$\pm$3.89**	97.66$\pm$1.48	**89.23$\pm$1.78**	**26.18$\pm$7.31**	98.09$\pm$0.29	43.69$\pm$1.49
HALX (ours)	86.78$\pm$0.08	81.56$\pm$0.07	98.20$\pm$0.13	89.11$\pm$0.07	27.21$\pm$0.14	98.25$\pm$0.05	48.27$\pm$0.50

higher false-positive rates. Inside HALX the XGBoost head lifts the BiLSTM encoder by 3.9 F1 points and cuts its FPR about 10-fold (from 7.80% to 0.77%). This confirms that on flow features the trees should make the final decision [10].

UNSW-NB15. On the official split all methods lose more than ten accuracy points. The test set has a different benign/attack mix (55% vs. 68% attacks), and its benign flows are evidently unlike those in training: at $\tau = 0.5$ every supervised model has a high FPR (27.18% for XGBoost, 27.21% for HALX). This well-documented gap [30], [31] is far more realistic than the near-perfect CICIDS2017 numbers. All supervised models then cluster around F1 $\approx$ 89%. HALX (89.11$_{\pm 0.07}$%) is tied with XGBoost (89.13$_{\pm 0.10}$%) and has the highest AUC among the deep models. The Transformer reaches a slightly higher mean F1 (89.23%), but is unstable: its standard deviation across seeds is 1.78 points and its FPR ranges from 18% to 36%. HALX has the most stable false-positive rate of all methods (std 0.14 points, compared with 0.33 for XGBoost). The XGBoost head again improves on the attention-free BiLSTM, by 1.3 F1 points and 7.5 macro-F1 points. On multi-class macro-F1, plain trees remain best on both datasets: the rare families (Worms, Infiltration; <0.1% of the data) gain little from the learned embedding.

Operating point. Fig. 2 shows that the ranking quality (AUC) of the tree-based models and HALX is much better than their $\tau = 0.5$ FPR suggests. At a 1% false-alarm budget on UNSW-NB15, HALX detects 86.9% of attacks. This is the highest of all methods: XGBoost reaches 85.9%, CNN-LSTM 84.9%, the Transformer 82.6% and Random Forest 77.7% (seed 0). In deployment, τ should be set from the SOC's alert budget.

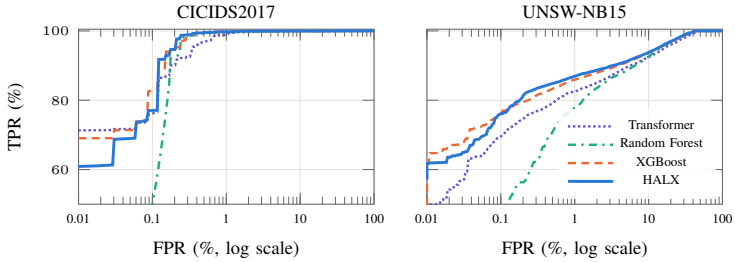


Fig. 2. ROC curves on the test sets (seed 0, log-scaled FPR). The low-FPR region corresponds to the alert budget of a SOC.

TABLE II

ABLATION OF HALX COMPONENTS (TEST F1, FPR AND MACRO-F1, %). CICIDS2017: 3 SEEDS; UNSW-NB15: SEED 0.

	CICIDS2017			UNSW-NB15		
Configuration	F1	FPR	M-F1	F1	FPR	M-F1
Full HALX	99.58$\pm$0.07	0.77$\pm$0.11	95.86$\pm$0.44	89.20	27.22	47.70
w/o attention (mean pooling)	99.63$\pm$0.07	0.69$\pm$0.11	97.26$\pm$0.95	88.89	27.38	48.44
w/o XGBoost (dense softmax head)	96.90$\pm$0.25	6.78$\pm$0.18	91.12$\pm$2.02	88.96	27.34	41.87
w/o raw features in head	99.41$\pm$0.10	1.10$\pm$0.11	93.75$\pm$1.21	88.93	27.13	45.01
w/o SMOTE	99.61$\pm$0.04	0.70$\pm$0.10	97.17$\pm$0.93	89.22	26.62	48.09
w/o privacy layer	99.74$\pm$0.02	0.48$\pm$0.12	95.43$\pm$0.95	89.17	26.89	49.15
BiLSTM only	96.11$\pm$0.93	6.71$\pm$1.36	84.88$\pm$2.97	87.50	33.13	39.82

B. Ablation Study

Table II isolates each component. The XGBoost head matters most: replacing it with the encoder's dense softmax head raises the CICIDS2017 FPR from 0.77% to 6.78% and lowers UNSW-NB15 macro-F1 from 47.70% to 41.87%. Passing the raw features to the head alongside the embedding also helps on both datasets (CICIDS2017 FPR 1.10% without them). Attention pooling, by contrast, does *not* improve accuracy on the saturated CICIDS2017 subset (F1 99.63% with mean pooling vs. 99.58%). It gives a small gain on UNSW-NB15 (88.89%→89.20%). Its main value is the per-alert explanation of Sec. V-D, obtained at no accuracy cost. SMOTE, common in

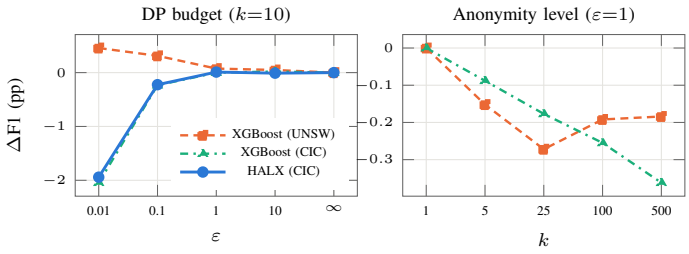


Fig. 3. Change in test F1 (percentage points, relative to the mechanism disabled) vs. DP budget ε (left, $k = 10$) and anonymity level k (right, $\varepsilon = 1$); seed 0.

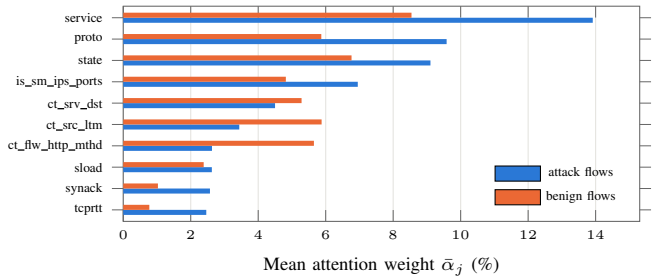


Fig. 4. Mean attention-pooling weight of the ten features most attended for attack flows (UNSW-NB15 test set, seed 0), compared with benign flows. A uniform weight would be $1/42 \approx 2.4\%$.

IDS pipelines [42], did not help either: binary F1 is unchanged without it and CICIDS2017 macro-F1 rises to 97.17%, so we recommend omitting it. The privacy layer costs 0.16 (CICIDS2017) and -0.03 (UNSW-NB15) F1 points.

C. Privacy–Utility Trade-off

Fig. 3 varies one mechanism at a time (XGBoost on both datasets; HALX on CICIDS2017, ε only). DP standardization is essentially free for $\varepsilon \geq 1$ ($|\Delta\text{F1}| \leq 0.07$). Only at $\varepsilon = 0.01$, where the noise added to each released mean has a standard deviation of 6.6 in g-space, does CICIDS2017 lose 2.0 F1 points, because the noisy statistics push many values to the clipping bounds. UNSW-NB15 has almost four times as many training flows, and since the noise shrinks as $1/n$ its F1 does not degrade even at $\varepsilon = 0.01$. Generalizing the quasi-identifiers costs 0.09–0.36 F1 points on CICIDS2017 for k from 5 to 500, and at most 0.27 on UNSW-NB15. The audited minimum equivalence-class size always met k (for example 515 at $k = 500$). HALX follows the same pattern (ΔF1 of -1.94, -0.22 and $+0.01$ at $\varepsilon = 0.01$, 0.1 and 1), so the default $\varepsilon = 1$, $k = 10$ is a sensible operating point.

D. Explainability and Severity Triage

Attention as explanation. Attention is far from uniform (Fig. 4). On UNSW-NB15 the ten most-attended tokens carry 57.8% of the weight of attack flows, compared with 23.8% under uniform weights. For attacks the encoder concentrates on service, protocol and connection state, on whether source and destination share IP and port (`is_sm_ips_ports`) and on handshake timing (`synack`, `tcprtt`). For benign flows the weight is spread more evenly over the connection-count features (`ct_*`). On CICIDS2017, attack flows emphasize the ACK-flag count and backward segment size, whereas benign flows put 25.8% of their attention on the FIN-flag count, consistent with gracefully closed TCP sessions. With each alert, HALX reports its top-attended features, which gives an analyst a concrete starting point without a post-hoc explainer.

Severity triage. The tiers are well ordered. On UNSW-NB15, 97.0% of the 42,156 high-severity alerts are true attacks, against 38.2% for medium and 22.8% for low severity, and 87.3% of all false positives fall into the low and medium tiers. An analyst who works through high-severity alerts first therefore sees mostly real attacks, even under the distribution shift that causes the high overall FPR. On CICIDS2017 the high tier reaches 99.8% precision.

E. Computational Cost

The HALX encoder has only 61,255 (CICIDS2017) and 61,690 (UNSW-NB15) parameters. Training on 2 CPU threads takes 11.0 and 13.9 minutes, respectively, including both XGBoost heads. In batch mode HALX scores 3,899 and 4,885 flows/s. Scoring a single flow takes a median of 3.74 ms (p95 4.88 ms) on UNSW-NB15, and the privacy layer adds about 0.73 μs per flow. No GPU is required.

VI. Discussion and Limitations

Scope of the privacy guarantees. HALX guarantees k-anonymity of the quasi-identifiers it stores, logs or forwards, and (ε, δ)-DP of the released normalization statistics. It does *not* make the encoder or the trees differentially private: a white-box adversary could still attempt membership inference on the weights. DP-SGD [29] or federated training across institutions [25], [26] would close this gap. k-anonymity also does not prevent attribute disclosure within an equivalence class [34].

Benchmarks vs. bank traffic. Neither benchmark is financial traffic, and CICIDS2017 models transfer poorly to other networks [31]. Our CICIDS2017 subset (2% of the flows) is near the ceiling for all tree-based models, so the shifted UNSW-NB15 split is more informative. Validation on bank flow logs is planned.

Temporal context. HALX scores flows independently. Multi-stage APT campaigns need cross-flow context (host-level session graphs or windows over timestamped flows), which the public partitions used here, lacking timestamps, cannot support without label leakage.

Adversarial robustness. We did not evaluate evasion attacks, which [32] lists among the evaluation gaps in ML-based security. We also did not test whether an attacker could manipulate the attention-based explanations; both are left to future work.

VII. Conclusion

We presented HALX, a hybrid Attention-LSTM and XGBoost detector for financial network flows. Its privacy layer

provides formally stated k-anonymity and differential-privacy guarantees. In a controlled comparison over three seeds, HALX matched the accuracy of the strongest tree ensembles and clearly outperformed CNN-LSTM, BiLSTM and Transformer detectors on CICIDS2017. Under the distribution shift of the official UNSW-NB15 split it detected the most attacks at a 1% false-alarm budget (seed-0 ROC analysis), and it had the most stable false-positive rate across seeds. The privacy layer cost at most 0.16 F1 points. The broader lesson for financial SOCs is that boosted trees should make the final decision on flow features. A deep encoder does not raise accuracy beyond a tuned XGBoost, but it adds per-alert explanations, a more stable false-alarm rate and a modest gain at low false-alarm budgets. Future work will add DP-SGD training, federated deployment across institutions, adversarial evaluation and validation on bank traffic.

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